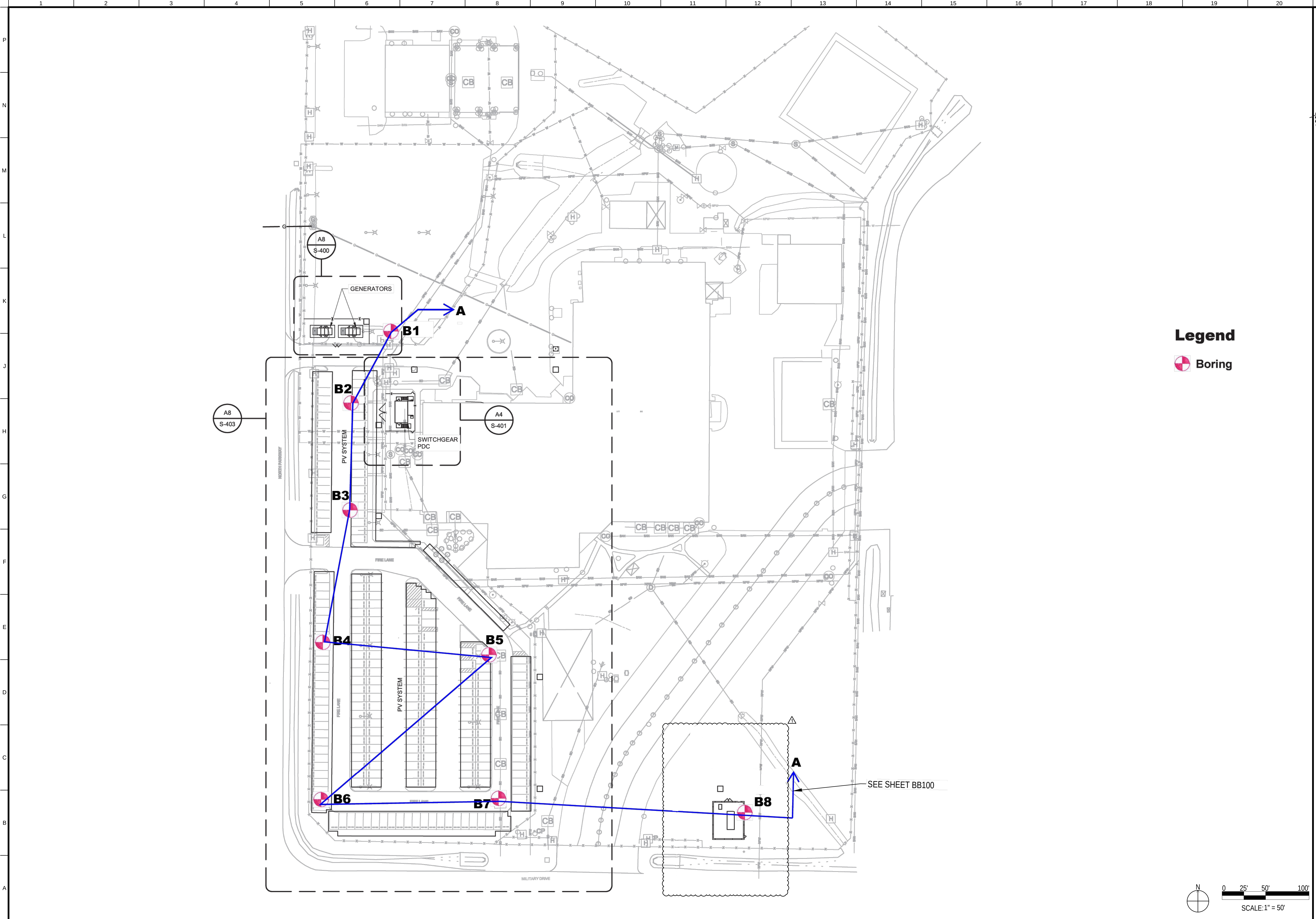


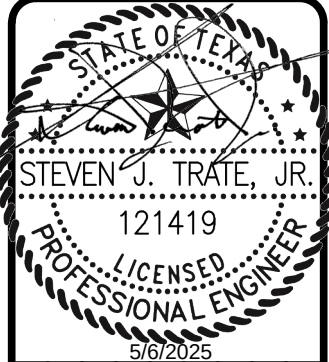


Legend

 Boring



US Army Corps
of Engineers®



STEVEN J. TRATE, J.
121419
LICENSED
PROFESSIONAL ENGINEER
9/22/25

MARK	DESCRIPTION	DATE
1	REVISED IN ACCORDANCE WITH AMENDMENT 002	5/6/2025

DESIGNED BY:
S. TRATE
24 JANUARY 2025

DRAWN BY:
K. KNOTH
15 JANUARY 2025

CHECKED BY:
K. KNOTH
15 JANUARY 2025

SUBMITTED BY:
C. DRUMMOND
15 JANUARY 2025

SIZE:
ANSI D

ISSUE DATE:
24 JANUARY 2025

SOLICITATION NO.:

CONTRACT NO.:

US ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
600 DR MARTIN LUTHER KING JR PL,
LOUISVILLE, KY 40202

MICHAEL BAKER INTERNATIONAL, INC.
100 AIRSIDE DRIVE
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CONROE ARMY RESERVE CENTER, CONROE, TEXAS
MICROGRID DESIGNS, 750KWDC

GEOTECHNICAL BORING PLAN

SHEET ID

B-100

CERTIFIED FINAL W912QR25RA023_Plan-0002

Plot Date: 12/30/2024 10:34:21 AM File Path: \\mnpa01a1.bur.mil.armcorp.com\VOL2\Projects\B874_Conroe_Arcc_Microgrid\05_Deliverables\Drawings\B-Civil\Sheet Plans\B874_L_G10a.dwg

MICROGRID OPERATIONAL DESCRIPTION

LOSS OF INCOMING UTILITY SOURCE

- 1. THE MICROGRID CONTROL SYSTEM (MGCS) AUTOMATICALLY ISOLATES ARC MICROGRID FROM THE UTILITY GRID (OPEN POI BREAKER(S) - ISLANDED MODE.
- 2. INITIATE GENERATOR UNIT(S) START AND SYNCHRONIZE THE GENERATOR UNIT(S) ON TO THE MICROGRID AND OPERATE IN PARALLEL WITH THE AVAILABLE DER ON THE MICROGRID.
- 3. SOLAR PV SYSTEM REMAIN CONNECTED AND IN OPERATION PROVIDING ENERGY INTO THE MICROGRID SYSTEM.
- 4. CONROE ARC FACILITIES REMAIN IN OPERATION IN ISLAND MODE

RETURN OF NORMAL UTILITY SUPPLY SOURCE

- 1. THE MGCS DETECTS THE RETURN OF HEALTHY UTILITY SERVICE SOURCE.
- 2.DISCONNECT AND DEENERGIZE ALL ONSITE GENERATION RESOURCE (GENERATOR AND PV).
- 3. MANUALLY CLOSE POI BREAKER TO THE HEALTHY UTILITY SOURCE.

INTENTIONAL ISLANDING OPERATION FROM GRID CONNECTED MODE

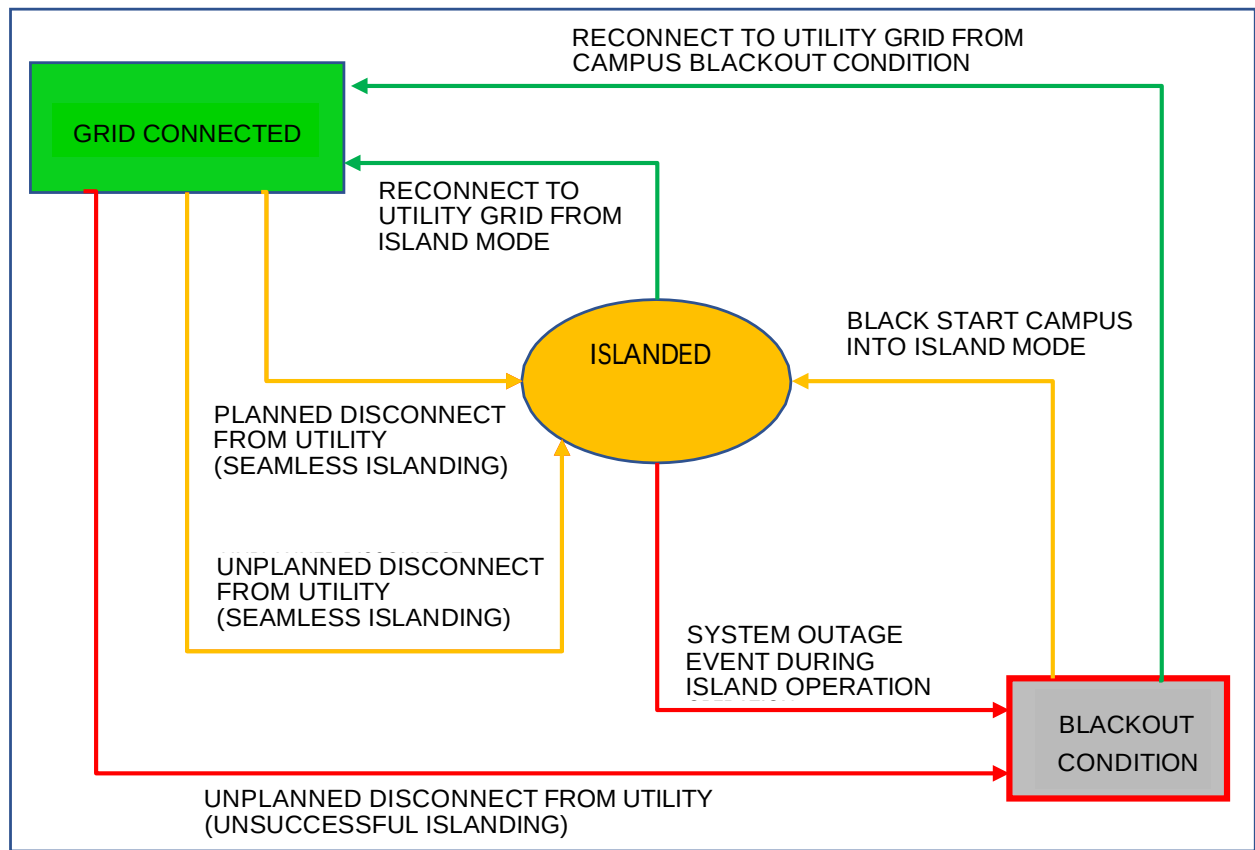
- 1. INITIATE START OF ALL GENERATOR UNITS AND BRING THE GENERATOR UNITS ONLINE TO OPERATE IN PARALLEL WITH THE UTILITY GRID AND PV.
- 2.DISPATCH DER (GENERATORS/PV) TO PICK UP ALL THE LOAD WHERE POWER FLOW IS NEAR ZERO (SLIGHTLY EXPORTING POWER) ACROSS POI.
- 3. OPEN POI BREAKER(S) TO ISOLATE ARC MICROGRID FROM THE UTILITY GRID FOR "SEAMLESS TRANSITION" TO ISLAND MODE.

THE MICROGRID CONTROL SYSTEM (MGCS) SHALL MANAGE THE OPERATION OF INTERCONNECTION SWITCHGEAR, DER, AND LOAD BALANCING FOR ALL MODES OF OPERATION WHEN THE MICROGRID CONTROL SYSTEM IS SET TO AUTOMATIC OPERATION.

MODE OF OPERATIONS

THERE ARE THREE STATES/MODES OF OPERATION FOR THE MICROGRID POWER SYSTEM: GRID CONNECTED, ISLANDED, AND BLACKOUT CONDITIONS. THE GRID CONNECTED AND THE ISLANDED MODES ARE THE TWO WORKING MODES OF OPERATIONS WHERE THE MICROGRID POWER SYSTEM IS ENERGIZED. IN BLACKOUT CONDITION, THE POWER SYSTEM IS COMPLETELY DEENERGIZED. SEE MICROGRID'S MODE OF OPERATION DIAGRAM BELOW FOR THE GRAPHICAL REPRESENTATION OF THE MICROGRID'S MODE OF OPERATION AND THE ASSOCIATED TRANSITION BETWEEN MODES OF OPERATION.

NOTE THAT FOR THIS PROJECT UNPLANNED SEAMLESS TRANSITION IS NOT POSSIBLE WITHOUT A BATTERY ENERGY STORAGE SYSTEM.



MICROGRID MODE OF OPERATION DIAGRAM

GRID CONNECTED MODE (NORMAL MODE OF OPERATION)

IN GRID CONNECTED MODE OPERATION, THE MICROGRID POWER SYSTEM IS CONNECTED IN PARALLEL TO THE UTILITY GRID INTERCONNECTION APPLICATION WILL BE FOR NET METERING. THE REAL AND REACTIVE POWER FLOW THROUGH THE POINT OF INTERCONNECTION (POI) WITH THE UTILITY COMPANY WITHIN OPERATING LIMITS. THE GRID CONNECTED MODE INCLUDES ACTIVE POWER LIMITS, REACTIVE POWER LIMITS, AND VOLTAGE LIMITS AT THE POI. THE MICROGRID CONTROL SYSTEM (MGCS) MANAGES THE AVAILABLE DISTRIBUTED ENERGY RESOURCES (DER) WHILE MAINTAINING THE OPERATING CONSTRAINTS OF THE MICROGRID POWER SYSTEM.

THE SOLAR PHOTOVOLTAIC SYSTEM GENERATES POWER WHEN SUNLIGHT IS AVAILABLE PROVIDING SUPPLEMENTAL POWER TO THE MICROGRID NETWORK. THE ON-SITE GENERATOR UNITS ARE NORMALLY OFF-LINE ON STANDBY. THE GENERATOR UNIT(S) WILL OPERATE WHEN REQUIRED DURING UTILITY POWER OUTAGE OR DURING THE DEMAND MANAGEMENT OR PEAK SHAVING PERIOD IF DESIRED.

ISLANDED MODE (CONTINGENCY MODE OF OPERATION)

THE MICROGRID POWER SYSTEM WILL AUTOMATICALLY TRANSITION FROM GRID CONNECTED MODE TO ISLAND MODE IN EVENT OF THE UTILITY GRID POWER OUTAGE. THE MICROGRID POWER SYSTEM WILL HAVE THE CAPABILITY TO TRANSITION FROM THE GRID CONNECTED MODE TO ISLAND MODE IN ANTICIPATION OF POTENTIAL OR PLANNED UTILITY POWER OUTAGE EVENTS SUCH AS SCHEDULED ROLLING BLACK, IMMINENT SEVERE WEATHER EVENTS, WILDFIRE EVENTS, ETC. THIS ACTION IS MANUALLY INITIATED BY THE SYSTEM OPERATOR VIA MGCS'S HMI.

IN ISLANDED MODE OF OPERATION, THE MICROGRID POWER SYSTEM UTILIZES THE AVAILABLE DERS ON THE CAMPUS SITE TO PROVIDE ELECTRICAL POWER TO THE CAMPUS FACILITIES.

THE SOLAR PHOTOVOLTAIC SYSTEM GENERATES POWER WHEN SUNLIGHT IS AVAILABLE PROVIDING POWER TO THE MICROGRID POWER SYSTEM. WHEN THE MICROGRID IS IN AN ISLANDED STATE, THE MGCS PROVIDES FREQUENCY CONTROL OR VOLTAGE CONTROL MODES:

- FREQUENCY CONTROL/REAL POWER SHARING -- THE MGCS WILL COMMAND ONE OF THE DERS TO OPERATE IN ISOCRONOUS MODE TO MAINTAIN THE MICROGRID SYSTEM FREQUENCY AT THE NOMINAL VALUE (60 HZ). THE MGCS WILL COMMAND THE REMAINING DERS TO OPERATE IN DROOP MODE. THE MGCS WILL ADJUST THE POWER OUTPUT SETPOINTS OF THESE DERS TO SHARE THE MICROGRID LOAD AND TO ENSURE THE ISOCRONOUS DER HAS ENOUGH OPERATING MARGIN TO RESPOND TO FREQUENCY DEVIATIONS.
- VOLTAGE CONTROL/REACTIVE POWER SHARING -- THE MGCS WILL COMMAND TWO GENERATORS TO OPERATE IN VOLTAGE DROOP MODE TO REGULATE THEIR TERMINAL VOLTAGES ACCORDING TO THEIR DROOP LINE. THE MGCS WILL ADJUST THE VOLTAGE SETPOINTS OF THESE DERS TO ENSURE THAT THEIR TERMINAL VOLTAGES DO NOT EXCEED PRESET LIMITS. THE MGCS WILL COMMAND THE REMAINING DERS TO OPERATE IN CONSTANT REACTIVE POWER MODE. THE MGCS WILL ADJUST THE REACTIVE POWER OUTPUT SETPOINTS OF THESE DERS TO ENSURE THAT THEIR TERMINAL VOLTAGES DO NOT EXCEED PRESET LIMITS. THE MGCS WILL ADJUST THE VOLTAGE SETPOINTS AND REACTIVE POWER SETPOINTS OF THE DERS TO SHARE THE MICROGRID REACTIVE POWER REQUIREMENT TO MAINTAIN SYSTEM VOLTAGE AT THE CONFIGURED VOLTAGE SETPOINT.

BLACKOUT CONDITION

IT IS POSSIBLE FOR THE MICROGRID POWER SYSTEM TO EXPERIENCE A COMPLETE BLACKOUT CONDITION AS A RESULT OF SEVERE FAULT ON THE POWER SYSTEM, MIS-OPERATION OR FAILURE OF THE MICROGRID CONTROL SYSTEM AND/OR PROTECTIVE RELAY SYSTEM. IN SUCH BLACKOUT EVENT, THE MICROGRID CONTROL SYSTEM WILL ISOLATE THE UTILITY GRID AND ALL OF THE DISTRIBUTED ENERGY RESOURCES FROM THE MICROGRID POWER SYSTEM. THE MICROGRID WILL HAVE THE CAPABILITY TO START ITS DER UNITS FROM THE DE-ENERGIZED STATE. THE MICROGRID POWER SYSTEM CAN BE BROUGHT BACK INTO SERVICE MANUALLY OR SEMI-AUTOMATICALLY AFTER THE POWER SYSTEM IS DEEMED IN SAFE CONDITION FOR REENERGIZATION.

WHEN THE MICROGRID IS IN A BLACKOUT STATE, THE MGCS COMMANDS THE DERS TO DISCONNECT FROM THE MICROGRID POWER SYSTEM. THE DERS REMAIN DISCONNECTED UNTIL A BLACK-START SEQUENCE IS INITIATED, OR THE MICROGRID IS RECONNECTED TO THE UTILITY GRID (SEE THE FOLLOWING TRANSITIONS SUBSECTION).

TRANSITIONAL STATES BETWEEN MODE OF OPERATION:

PLANNED DISCONNECT FROM THE UTILITY GRID

PLANNED INTENTIONAL ISLANDING INVOLVES CONTROLLING THE ENERGY FLOW AT THE POINT OF INTERCONNECTION(POI) TO NEAR ZERO AND SEPARATING FROM THE GRID. DURING PLANNED INTENTIONAL ISLANDING A MICROGRID DISCONNECTS FROM THE AREA ELECTRIC POWER SYSTEM (EPS) IN A PLANNED MANNER WHEN THE EPS IS GRID-CONNECTED AND IN A NORMAL OPERATING MODE. THE TRANSITION TO ISLAND MODE IS BASED ON A REASONABLE WARNING PERIOD.

FOR THE PLANNED DISCONNECT FROM THE UTILITY GRID, THE MICROGRID SYSTEM OPERATOR WILL INITIATE GOTO ISLAND MODE. THE MGCS WILL RESPOND TO A COMMAND TO DISCONNECT THE MICROGRID FROM THE UTILITY BY DISPATCHING THE DERS AND OPENING THE MICROGRID POI. PRIOR TO OPENING THE POI, THE MGCS WILL EXECUTE CONTROL ACTIONS TO PREPARE THE MICROGRID FOR A SUCCESSFUL TRANSITION TO ISLAND OPERATION AND TO MITIGATE THE NEED TO SHED LOAD.

THE MGCS WILL PERFORM THE FOLLOWING FUNCTIONS DURING A PLANNED SEAMLESS DISCONNECT:

1. VERIFIES PERMISSIVE CONDITIONS OF THE MICROGRID POWER SYSTEM TO INITIATE GOTO ISLAND MODE (DISCONNECT FROM THE UTILITY GRID) PROCESS.
2. STARTS BOTH STANDBY GENERATOR UNITS TO PROVIDE ADDITIONAL SPINNING RESERVE AND INERTIA ON THE MICROGRID.
3. ADJUSTS RENEWABLE DER POWER OUTPUTS FOR ISLAND OPERATION
4. DISPATCHES GENERATORS TO REDUCE POWER FLOW (REAL AND REACTIVE) ACROSS THE MICROGRID POI
5. OPENS THE POI BREAKER WHEN THE REAL AND REACTIVE POWER FLOWS ARE WITHIN DISCONNECTS LIMITS. [THE REAL AND REACTIVE POWER FLOWS SHOULD BE NEAR ZERO, BUT SLIGHTLY IMPORTING REAL AND REACTIVE POWER.]
6. AUTOMATICALLY CHANGES ONE DER (GENERATOR UNIT) TO ISOCRONOUS MODE
7. THE SOLAR PHOTOVOLTAIC SYSTEM WILL AUTOMATICALLY DISCONNECT ITSELF FORM THE MICROGRID THROUGH THE INVERTER WHEN IT SENSE MICROGRID INSTABILITY (VOLTAGE AND/OR FREQUENCY ARE OUTSIDE OF ACCEPTABLE LIMITS) DURING THE TRANSITION OR WHEN THERE IS INSUFFICIENT SUNLIGHT ENERGY AVAILABLE TO PRODUCE POWER. THE SOLAR PHOTOVOLTAIC SYSTEM WILL AUTOMATICALLY SWITCH MODE OF OPERATION FROM ISLAND MODE TO GRID CONNECTED MODE. THE PV SYSTEM WILL AUTOMATICALLY RECONNECT ITSELF TO THE MICROGRID WHEN THE MICROGRID POWER SYSTEM IS HEALTHY AND STABLE (AT NORMAL OPERATING VOLTAGE AND FREQUENCY) AND WHEN SUNLIGHT IS AVAILABLE TO PRODUCE POWER AND INJECT TO POWER INTO THE MICROGRID.

UNPLANNED DISCONNECT FROM THE UTILITY GRID (NON-SEAMLESS)

UNPLANNED ISLANDING WILL OCCUR WHEN THE MICROGRID PROTECTIVE RELAY SYSTEM DETECTS THE SUDDEN LOSS OF UTILITY VOLTAGE OR ABNORMAL UTILITY VOLTAGE, CURRENT AND/OR FREQUENCY BEYOND ACCEPTABLE LIMITS. THE DETECTION OF UTILITY SOURCE ABNORMAL CONDITIONS WILL INITIATE THE MGCS TO TRANSITION TO ISLANDED MODE BY DISPATCHING THE DERS AND OPENING THE MICROGRID POI. THE MGCS WILL PERFORM THE FOLLOWING FUNCTIONS IN RESPONSE TO DETECTION OF UNACCEPTABLE UTILITY SOURCE ABNORMAL CONDITION:

1. POI RELAY OPENS THE POI BREAKER
2. THE SOLAR PHOTOVOLTAIC SYSTEM WILL AUTOMATICALLY DISCONNECT ITSELF FORM THE MICROGRID THROUGH THE INVERTER WHEN IT SENSE MICROGRID INSTABILITY (VOLTAGE AND/OR FREQUENCY ARE OUTSIDE OF ACCEPTABLE LIMITS) DURING THE TRANSITION OR WHEN THERE IS INSUFFICIENT SUNLIGHT ENERGY AVAILABLE TO PRODUCE POWER.
3. OPEN ALL LOAD BREAKERS ON ARC MAIN DISTRIBUTION PANEL POWER PANEL PP-1 LOADS TO UNLOAD SYSTEM FOR RESTART.
4. START STANDBY GENERATOR UNITS
5. THE PV SYSTEM WILL AUTOMATICALLY RECONNECT ITSELF TO THE MICROGRID WHEN THE MICROGRID POWER SYSTEM IS HEALTHY AND STABLE (AT NORMAL OPERATING VOLTAGE AND FREQUENCY) AND WHEN SUNLIGHT IS AVAILABLE TO PRODUCE POWER AND INJECT TO POWER INTO THE MICROGRID.
6. CLOSE ARC BUILDING FEEDER BREAKERS ONE BY ONE TO ENERGIZE FACILITY LOADS.

THERE ARE SCENARIOS WHERE THE MICROGRID PROTECTIVE SYSTEM MAY NOT ONLY TRIP THE POI BREAKER(S), BUT ALSO TRIP THE DERS RESULTING IN MICROGRID BLACKOUT CONDITION. IF SUCH AN EVENT OCCURRED, THE MGCS WILL DISCONNECT ALL DERS FROM THE DEAD MICROGRID POWER SYSTEM. THE MGCS WILL ALSO RESET THE DER'S SETPOINTS PARAMETERS TO PREPARE FOR A MANUAL OR AUTOMATIC BLACK-START OPERATION. THE MICROGRID CONTROL SYSTEM WILL BE POWERED FROM THE UNINTERRUPTIBLE POWER SUPPLY (UPS) SOURCES. ONCE THE MICROGRID IS BLACKED OUT CONDITION, THE MICROGRID WILL HAVE THE ABILITY TO BE RE-ENERGIZED BY ONE OF THE FOLLOWING METHODS:

1. FOR MOMENTARY OUTAGE (LESS THAN ONE MINUTE):
 - a. THE SOLAR PHOTOVOLTAIC SYSTEM WILL AUTOMATICALLY DISCONNECT ITSELF FORM THE MICROGRID THROUGH THE INVERTER WHEN IT SENSE MICROGRID INSTABILITY (VOLTAGE AND/OR FREQUENCY ARE OUTSIDE OF ACCEPTABLE LIMITS) DURING THE TRANSITION OR WHEN THERE IS INSUFFICIENT SUNLIGHT ENERGY AVAILABLE TO PRODUCE POWER. THE SOLAR PHOTOVOLTAIC SYSTEM WILL AUTOMATICALLY SWITCH MODE OF OPERATION FROM ISLAND MODE TO GRID CONNECTED MODE.
 - b. THE PV SYSTEM WILL AUTOMATICALLY RECONNECT ITSELF TO THE MICROGRID WHEN THE MICROGRID POWER SYSTEM IS HEALTHY AND STABLE (AT NORMAL OPERATING VOLTAGE AND FREQUENCY) AND WHEN SUNLIGHT IS AVAILABLE TO PRODUCE POWER AND INJECT TO POWER INTO THE MICROGRID.
 - c. FACILITY LOADS RETURN ONCE UTILITY RETURNS TO NORMAL POWER.
2. FOR TEMPORARY OR LONGER DURATION OF UTILITY POWER OUTAGE, THE ON-SITE GENERATOR UNITS ARE BROUGHT ONLINE TO PROVIDE ELECTRICAL POWER TO THE MICROGRID SYSTEM TO SERVE THE CAMPUS LOADS IN ISLAND MODE UNTIL THE NORMAL UTILITY POWER SOURCE BECOME AVAILABLE. THE PV SYSTEM WILL AUTOMATICALLY DISCONNECT AND RECONNECT ITSELF AS DESCRIBED IN ITEM 1. THE OPERATOR MAY RE-ENERGIZE THE MICROGRID BY INITIATING THE MGCS BLACK-START SEQUENCE.

A QUALIFIED SYSTEM OPERATOR NEEDS TO INVESTIGATE AND EXAMINE THE STATE OF CONDITIONS OF THE MICROGRID POWER SYSTEM BEFORE RE-ENERGIZATION OF THE MICROGRID POWER SYSTEM AFTER A BLACKOUT EVENT. THIS IS TO ENSURE THAT ANY DANGEROUS CONDITIONS ARE IDENTIFIED, AND THE SYSTEM IS PLACED BACK IN A SAFE CONDITION FOR RE-ENERGIZATION.

RECONNECT THE DEAD MICROGRID TO THE UTILITY GRID (BLACKOUT STATE TO GRID CONNECTED MODE)

THE MICROGRID SYSTEM OPERATOR INITIATES GOTO GRID CONNECTED MODE FROM THE MGCS'S HMI(S) OR THE AUTOMATIC RECONNECT FUNCTION OF THE MGCS. THE MGCS WILL PERFORM THE FOLLOWING FUNCTIONS TO RECONNECT THE LIVE MICROGRID TO THE UTILITY GRID:

1. CONFIRM HEALTHY UTILITY SOURCE (VOLTAGE AND FREQUENCY ARE WITHIN ACCEPTABLE AND STABLE LIMITS)
2. DISCONNECT AND DEENERGIZE ALL ONSITE GENERATION RESOURCE (GENERATOR AND PV)
3. CLOSE THE MICROGRID POI BREAKER

NOTE: THIS IS THE FASTEST WAY TO RESTORE POWER TO THE MICROGRID AND TO RESTORE ELECTRICAL SUPPLY SOURCE TO THE CAMPUS AFTER A BLACKOUT. THE DERS CAN REMAIN OFF-LINE ALLOWING THE SYSTEM OPERATOR TIME TO ACCESS THE CONDITION OF DERS AFTER A BLACKOUT BEFORE BRINGING THEM BACK ON-LINE.

BLACK START MICROGRID

THE MICROGRID SYSTEM OPERATOR INITIATES BLACKSTART FROM THE MGCS'S HMI(S). THE MGCS WILL INITIATE BLACK START SEQUENCE UNDER ITS OWN POWER USING AVAILABLE DERS. THIS METHOD WILL ALLOW THE MICROGRID TO BE RESTORED AND OPERATED DURING A UTILITY OUTAGE. THE MGCS WILL PERFORM THE FOLLOWING FUNCTIONS TO BLACK START MICROGRID POWER SYSTEM:

1. CONFIRM THAT THE MICROGRID POWER SYSTEM IS DEAD (SYSTEM VOLTAGE IS BELOW THE MICROGRID DEAD THRESHOLD).
2. CONFIRM THAT THE POI BREAKER IS OPENED AND ISOLATED FROM THE UTILITY GRID AND ALL DERS ARE DISCONNECTED FROM THE MICROGRID ISLAND.
3. OPEN FACILITY LOAD FEEDER BREAKERS AT THE ARC MAIN DISTRIBUTION PANEL AND POWER PANEL PP-1 LOADS.
4. STARTS BOTH STANDBY GENERATOR UNITS
5. ENERGIZE THE MAIN UTILITY SERVICE SWITCHGEAR BY CLOSING THE GENERATOR UNIT TIE BREAKER(S)
6. ENERGIZE DISTRIBUTION ARC MAIN DISTRIBUTION PANEL AND POWER PANEL PP-1 LOADS BY CLOSING BREAKERS ONE AT A TIME.
7. THE PV SYSTEM WILL AUTOMATICALLY RECONNECT ITSELF TO THE MICROGRID WHEN THE MICROGRID POWER SYSTEM IS HEALTHY AND STABLE (AT NORMAL OPERATING VOLTAGE AND FREQUENCY) AND WHEN SUNLIGHT IS AVAILABLE TO PRODUCE POWER AND INJECT TO POWER INTO THE MICROGRID.

NOTE THAT THIS IS THE FASTEST WAY TO RESTORE POWER TO THE MICROGRID AND TO RESTORE POWER TO THE CAMPUS AFTER A BLACKOUT. THE DERS CAN REMAIN OFF-LINE ALLOWING THE SYSTEM OPERATOR TIME TO ASSESS THE CONDITION OF DERS AFTER A BLACKOUT BEFORE BRINGING THEM BACK ON-LINE.

PV INVERTER OPERATION DESCRIPTION

- A. MANUAL POWER ON/OFF - NORMALLY, IT IS NOT NECESSARY TO TURN OFF THE INVERTER, BUT IT CAN BE SHUT DOWN MANUALLY IF REGULATION SETTING OR MAINTENANCE IS REQUIRED. RESTARTING THE INVERTER IS REQUIRED AFTER REGULATION SETTING OR MANUAL (FAULT) SHUTDOWN. THE INVERTER WILL START UP AND OPERATE NORMALLY IF THE START-UP CONDITIONS ARE MET. OTHERWISE, THE INVERTER WILL GO TO STAND-BY MODE.
- B. AUTOMATIC TURN ON/OFF - THE INVERTER WILL START UP AUTOMATICALLY WHEN THE OUTPUT VOLTAGE AND POWER OF PV ARRAYS MEET THE SET VALUE. AC POWER GRID IS NORMAL AND THE AMBIENT TEMPERATURE IS WITHIN ALLOWABLE OPERATING RANGE. THE INVERTER WILL BE SHUT DOWN AUTOMATICALLY WHEN THE OUTPUT VOLTAGE AND POWER OF PV MODULES ARE LOWER THAN THE SET VALUE, THE AC POWER GRID FAILS, OR THE AMBIENT TEMPERATURE EXCEEDS THE NORMAL RANGE
- C. OPERATION MODES - THERE ARE 4 OPERATION MODES: LOGO/START-UP, NORMAL OPERATION, STANDBY AND FAULT MODES.
 1. START-UP MODE IS WHEN THE INVERTER IS CHECKING IF CONDITIONS ARE MET FOR CONNECTION AFTER A MANUAL START-UP OF THE INVERTER.
 2. NORMAL OPERATION MODE IS THE DEFAULT INDICATION INTERFACE FOR NORMAL OPERATION. IN THIS MODE, THE INVERTER IS CONVERTING THE DC POWER FROM THE ARRAY TO AC POWER, FEEDING IT CONTINUOUSLY TO THE GRID.
 3. STANDBY MODE IS ACTIVE WHEN THE OUTPUT VOLTAGE AND POWER OF PV MODULES DO NOT MEET THE STARTUP CONDITIONS OR PV VOLTAGE AND INPUT POWER ARE LOWER THAN THE SET VALUE. THE INVERTER WILL CHECK AUTOMATICALLY WHETHER IT MEETS THE STARTUP CONDITIONS IN THIS MODE UNTIL IT TURNS BACK TO NORMAL OPERATION MODE.
 4. FAULT MODE IS DISPLAYED WHEN THE INVERTER DISCONNECTS FROM THE GRID DUE TO A FAULT IN THE INVERTER OR GRID.
- D. GRID-TIED POWER GENERATION -THE INVERTERS WILL HAVE AN AUTOMATIC GRID-TIED POWER GENERATION PROCESS, WHICH WILL CHECK CONSTANTLY WHETHER AC POWER GRID MEETS THE CONDITIONS FOR GRID-TIED POWER GENERATION, AND TEST WHETHER THE PV ARRAY HAS ADEQUATE ENERGY. AFTER ALL CONDITIONS ARE MET, THE INVERTER WILL ENTER NORMAL OPERATION MODE AND ENTER NET METERING WITH THE GRID. WHILE GENERATING POWER, THE INVERTER CAN DETECT THE POWER GRID AT ALL TIMES, SUPPLY CURRENT, MATCHING VOLTAGE AND FREQUENCY OF THE GRID AND KEEP THE PHOTOVOLTAIC ARRAY OUTPUT AT THE MAXIMUM POWER POINT.

STANDBY GENERATOR UNIT SYNCHRONIZATION CABILITY

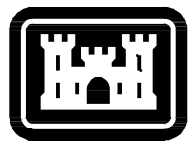
- A. EACH OF THE STANDBY GENERATOR UNIT SHALL HAVE THE CAPABILITY TO SYNCHRONIZE TO THE MICROGRID ACROSS THEIR RESPECTIVE GENERATOR TIE BREAKER, OR AT THE UTILITY POI BREAKER.
- B. EACH OF THE STANDBY DIESEL GENERATOR UNIT SHALL HAVE THE CAPABILITY TO OPERATE IN THE "MICROGRID CONTROL MODE" OR IN THE "LOCAL CONTROL MODE" WHICH IS SELECTED BY LOCAL CONTROL SELECTOR SWITCH LOCATED ON THE RESPECTIVE GENERATOR CONTROL CUBICLE ON THE MV SWITCHGEAR.
- C. EACH GENERATOR UNIT SHALL HAVE ITS OWN DEDICATED AUTO-SYNCHRONIZER AND CAN OPERATE INDEPENDENTLY FROM THE MICROGRID CONTROLLER WHEN PLACED IN THE "LOCAL CONTROL MODE". THE LOCAL GENERATOR CONTROL/AUTO-SYNCHRONIZER WILL PERFORM THE GENERATOR UNIT STARTER/STOP COMMANDS, AVR RAISE/LOWER COMMAND, GOVERNOR CONTROL, SYNCHRONIZATION, SYNCH SOURCE SELECTION AND APPROPRIATE BREAKER CLOSE/OPEN/TRIP COMMANDS. THE MICROGRID CONTROLLER SHALL EXECUTE THE CONTROL AND MONITORING FUNCTIONS THROUGH THE LOCAL GENERATOR CONTROL/AUTO-SYNCHRONIZER.
- D. IN THE "MICROGRID CONTROL MODE", THE MICROGRID CONTROLLER EXECUTES THE AUTOMATIC GENERATOR STARTER/STOP COMMANDS, SYNCHRONIZATION, SYNCH SOURCE SELECTION AND APPROPRIATE BREAKER CLOSE/OPEN/TRIP COMMANDS BASED ON PROGRAMMED SEQUENCE OF OPERATION.
- E. IN THE "LOCAL CONTROL MODE", THE OPERATOR WILL NEED TO MANUALLY INITIATE THE GENERATOR STARTER/STOP COMMANDS, SYNCH SOURCE SELECTION VIA SOURCE SELECTOR SWITCH (GEN BREAKER/POI BREAKER), SYNCHRONIZATION, AND BREAKER CLOSE COMMANDS BASED ON ESTABLISHED SEQUENCE OF MANUAL GENERATOR SYNCHRONIZATION OPERATION.

LOAD BANK TESTING SEQUENCE:

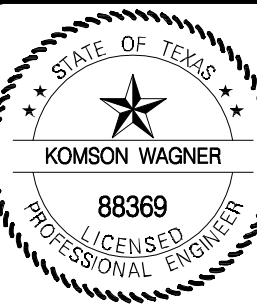
- A. CONNECT PORTABLE STEPPED LOAD BANK TO OPEN LOAD BANK DISCONNECT. TRANSFER ALL LOADS OFF OF SWITCHGEAR BUS 2 AND OPEN BREAKERS CB-F1A, CBF2A, AND TIE BREAKER.
- B. CLOSE BREAKER CB-F4 AND LOAD BREAK DISCONNECT SWITCH.
- C. CLOSE BREAKER CB-G1 AND TURN ON GENERATOR 1.
- D. STEP PORTABLE LOAD BANK UP SEQUENTIALLY UP TO FULL LOAD THEN REDUCE LOAD STEPPING BACK DOWN TO NO LOAD.
- E. TURN GENERATOR 1 OFF AND OPEN BREAKER CB-G1.
- F. REPEAT STEPS C THROUGH E WITH GENERATOR 2 IF CLIN0003 IS SELECTED.
- G. OPEN BREAKER CB-F4 AND LOAD BREAK DISCONNECT SWITCH.
- H. CLOSE TIE BREAKERS AND RETURN BREAKERS CB-F1A AND CB-F2A BACK TO PREVIOUS POSITION.
- I. DISCONNECT AND REMOVE PORTABLE LOAD BANK.

CLIN NOTES:

- CLIN 0003: ADD GEN2 340KW (AT .8PF) PRIME RATED GENERATOR.
- CLIN 0004: ADD PV/CARPORT #4 (61KW)
- CLIN 0005: ADD PV/CARPORT #5 (63.44KW)
- CLIN 0006: ADD PV/CARPORT #6 (102.48KW)
- CLIN 0007: ADD PV/CARPORT #7 (94.85KW)
- CLIN 0008: ADD PV/CARPORT #8 (101.87KW)
- CLIN 0009: ADD ELECTRICAL SERVICE FROM POWER PANEL PP-1 TO EXISTING FIRE HOUSE MAIN SERVICE ENTRANCE RATED DISCONNECT.



US Army Corps of Engineers



05/06/25

DATE	DESCRIPTION	MARK
05/06/25	AMENDMENT 0002	1

DESIGNED BY: M. WAGNER	ISSUE DATE: 06 MAY 2025
DRAWN BY: J. ROGERS	SOLICITATION NO.:
CHECKED BY: M. SMITH	CONTRACT NO.:
SUBMITTED BY: C. DRUMMOND	
SIZE: ANSI	

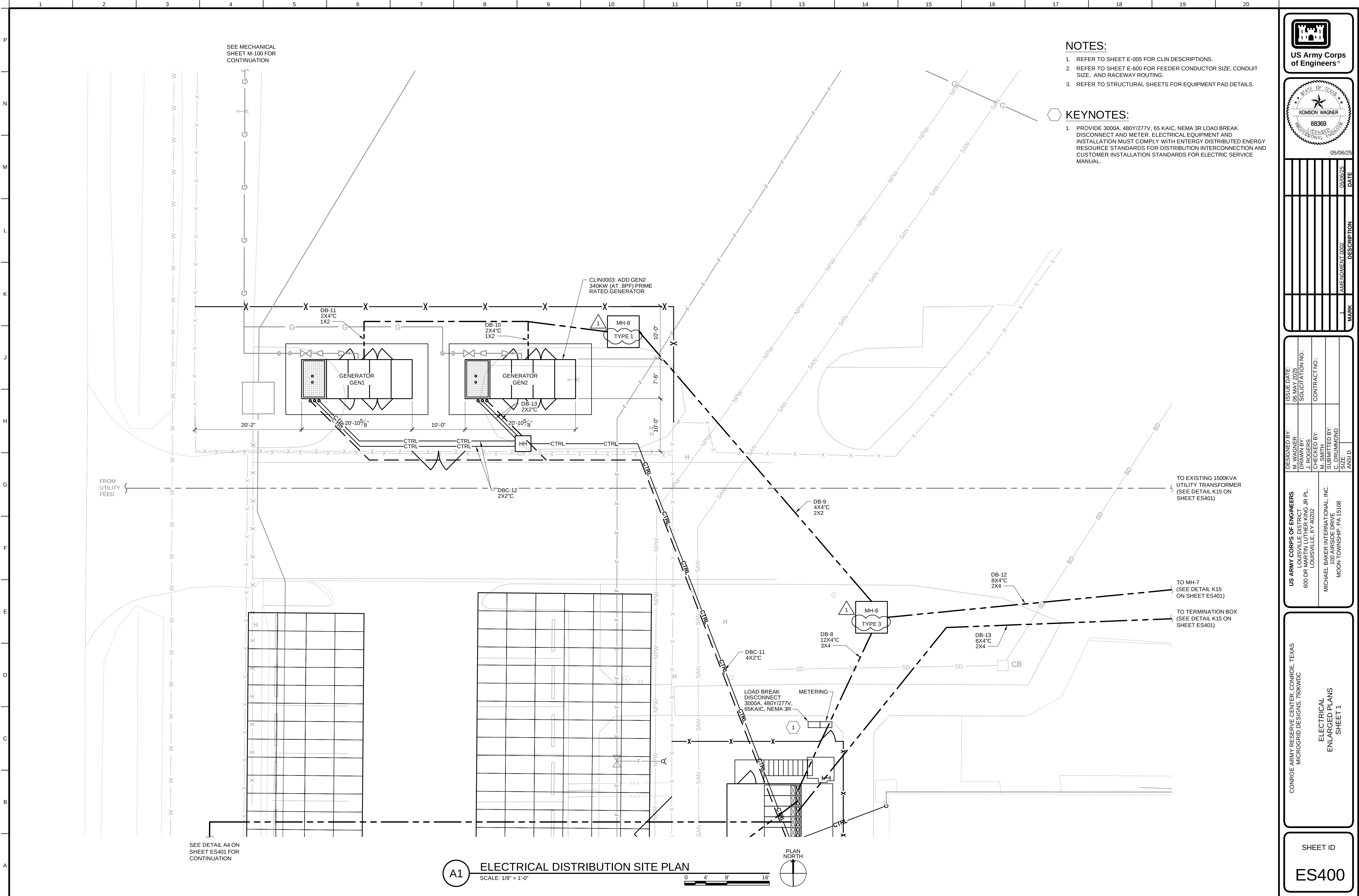
US ARMY CORPS OF ENGINEERS	LOUISVILLE DISTRICT
600 DR MARTIN LUTHER KING JR PL,	LOUISVILLE, KY 40202
MICHAEL BAKER INTERNATIONAL, INC.	1500 N. BROADWAY
MOON TOWNSHIP, PA 15108	

CONROE ARMY RESERVE CENTER, CONROE, TEXAS
MICROGRID DESIGNS, 750KWD

ELECTRICAL
MICROGRID OPERATION DESCRIPTION

SHEET ID

E-005



- NOTES:**
- REFER TO SHEET E-005 FOR CLIN DESCRIPTIONS.
 - REFER TO SHEET E-600 FOR FEEDER CONDUCTOR SIZE, CONDUIT SIZE, AND RACEWAY ROUTING.
 - REFER TO STRUCTURAL SHEETS FOR EQUIPMENT PAD DETAILS.

- KEYNOTES:**
- PROVIDE 3000A, 480Y/277V, 65 KAIC, NEMA 3R LOAD BREAK DISCONNECT AND METER, ELECTRICAL EQUIPMENT AND INSTALLATION MUST COMPLY WITH ENTERGY DISTRIBUTED ENERGY RESOURCE STANDARDS FOR DISTRIBUTION INTERCONNECTION AND CUSTOMER INSTALLATION STANDARDS FOR ELECTRIC SERVICE MANUAL.

US Army Corps of Engineers

05/06/25

MARK	AMENDMENT	DESCRIPTION	DATE
1	002		05/06/25

DESIGNED BY: M. WAGNER	ISSUE DATE: 06 MAY 2025
DRAWN BY: J. ROGERS	SOLICITATION NO.:
CHECKED BY: M. SMITH	CONTRACT NO.:
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US ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
600 DR MARTIN LUTHER KING JR PL,
LOUISVILLE, KY 40202

MICHAEL BAKER INTERNATIONAL, INC.
1000 RIVERCHASE DRIVE
MOON TOWNSHIP, PA 15108

CONROE ARMY RESERVE CENTER, CONROE, TEXAS
MICROGRID DESIGNS, 750KWDC

ELECTRICAL
ENLARGED PLANS
SHEET 1

SHEET ID

ES400

A1

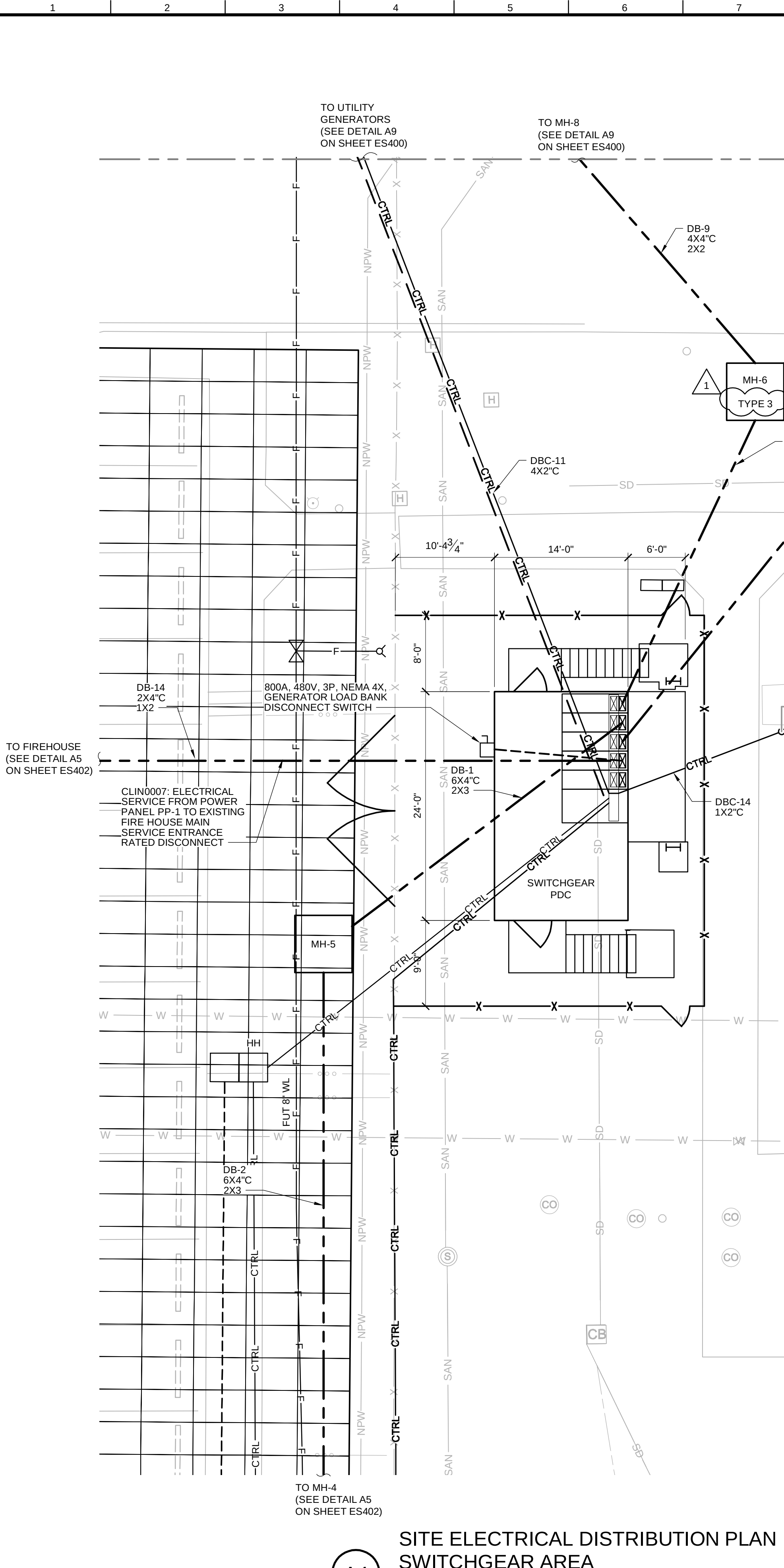
ELECTRICAL DISTRIBUTION SITE PLAN
SCALE: 1/8" = 1'-0"

0 4' 8' 16'

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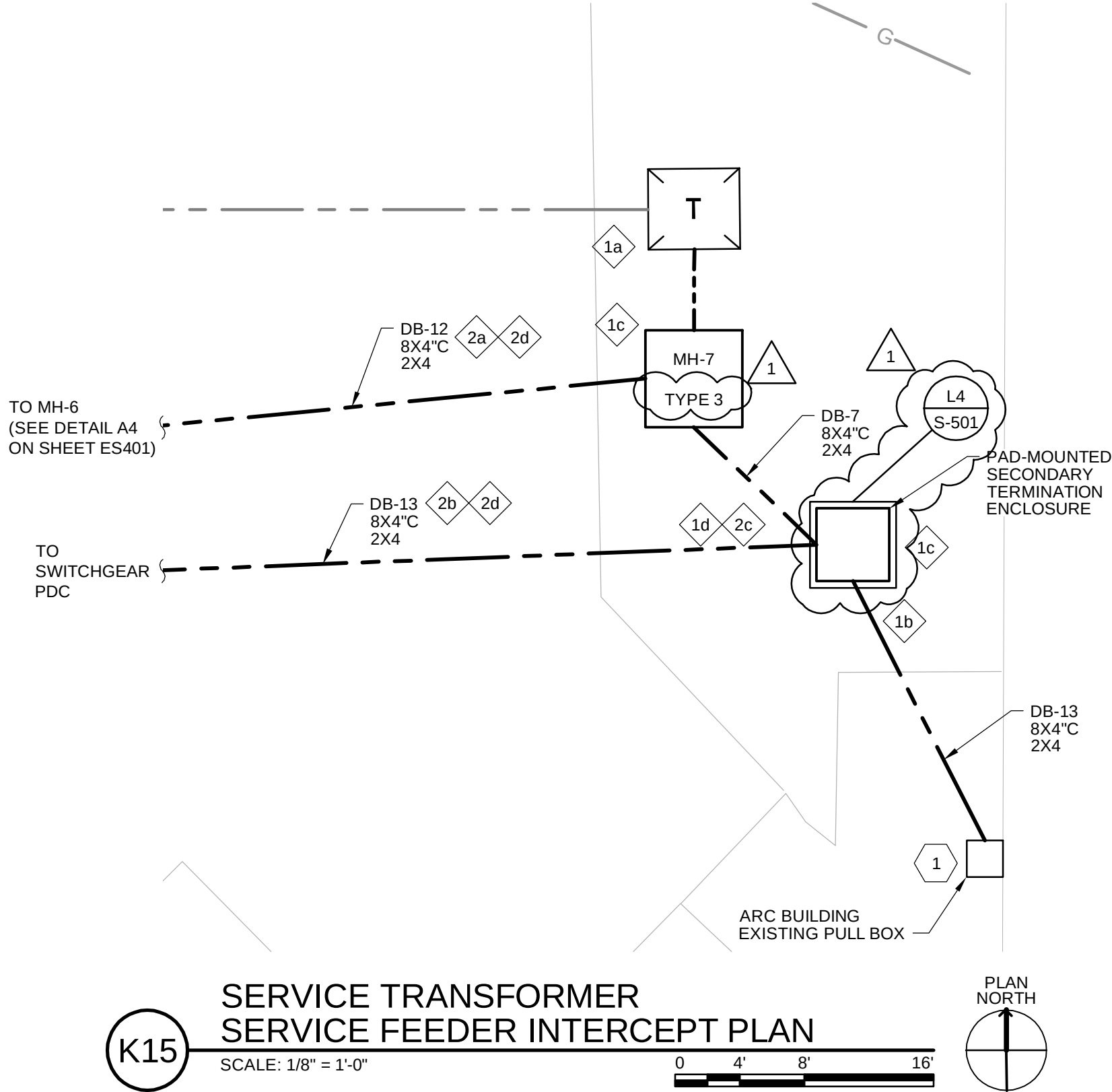
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**SITE ELECTRICAL DISTRIBUTION PLAN
SWITCHGEAR AREA**

SCALE: 1/8" = 1'-0"



**SERVICE TRANSFORMER
SERVICE FEEDER INTERCEPT PLAN**

SCALE: 1/8" = 1'-0"



NOTES:

1. REFER TO SHEET E-005 FOR CLIN DESCRIPTIONS.
2. REFER TO SHEET E-600 FOR FEEDER CONDUCTOR SIZE, CONDUIT SIZE, AND RACEWAY ROUTING.
3. REFER TO STRUCTURAL SHEETS FOR EQUIPMENT PAD DETAILS.
4. PROVIDE TEMPORARY POWER AS REQUIRED TO MAINTAIN ARC BUILDING OPERATIONS THROUGHOUT CONSTRUCTION.
5. ROUTE 2" CONDUIT ALONG SIDE OF BUILDING, AVOIDING INTERFERENCES ALONG THE NORTH SIDE OF THE BUILDING. FOR CONTINUATION SEE ES402.

**SUGGESTED CONSTRUCTION
PHASING:**

1. PHASE 1:
 - a. COORDINATE WITH OWNER AND ENTERGY PRIOR TO ANY SHUTDOWNS. FIELD VERIFY EXISTING CONDITIONS AND LOCATE UTILITIES PRIOR TO BEGINNING WORK. FINAL LOCATION OF ELECTRICAL EQUIPMENT SHALL BE FIELD LOCATED.
 - b. INTERCEPT SECONDARY FEED FROM EXISTING 1500KVA TRANSFORMER TO ARC BUILDING PULLBOX.
 - c. PROVIDE NEW MANHOLE AND 3000A, 600V, 3P, 4W, NEMA 3R PAD-MOUNTED SECONDARY TERMINATION ENCLOSURE. PAD-MOUNTED SECONDARY TERMINATION ENCLOSURE MUST BE LOCATED IN LINE WITH INTERCEPTED ARC BUILDING FEEDER.
 - d. PROVIDE NEW CONDUCTORS, CONDUIT, AND DUCTBANK FROM TRANSFORMER SECONDARY THROUGH MANHOLE AND TERMINATE INSIDE PAD-MOUNTED SECONDARY TERMINATION ENCLOSURE. TERMINATE INTERCEPTED ARC BUILDING FEEDER INSIDE PAD-MOUNTED SECONDARY TERMINATION ENCLOSURE.
2. PHASE 2:
 - a. PROVIDE CONDUCTORS, CONDUIT, AND DUCTBANK FROM TRANSFORMER SECONDARY TO MICROGRID SWITCHGEAR BREAKER M1-POI.
 - b. PROVIDE CONDUCTORS, CONDUIT, AND DUCTBANK FROM MICROGRID SWITCHGEAR BREAKER CB-F1 TO PAD-MOUNTED SECONDARY TERMINATION ENCLOSURE.
 - c. REMOVE CONDUCTORS FROM TRANSFORMER SECONDARY TO PAD-MOUNTED SECONDARY TERMINATION ENCLOSURE.
 - d. TRANSITION POWER TO PAD-MOUNTED SECONDARY TERMINATION ENCLOSURE FROM TRANSFORMER TO MICROGRID SWITCHGEAR.

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AMENDMENT 002
DESCRIPTION

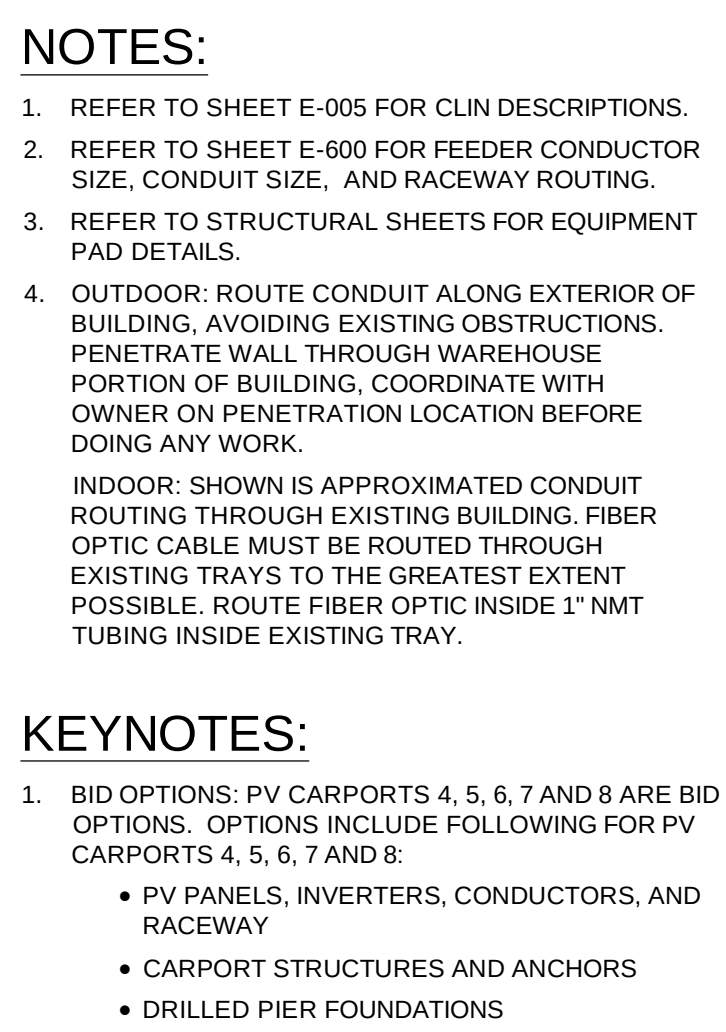
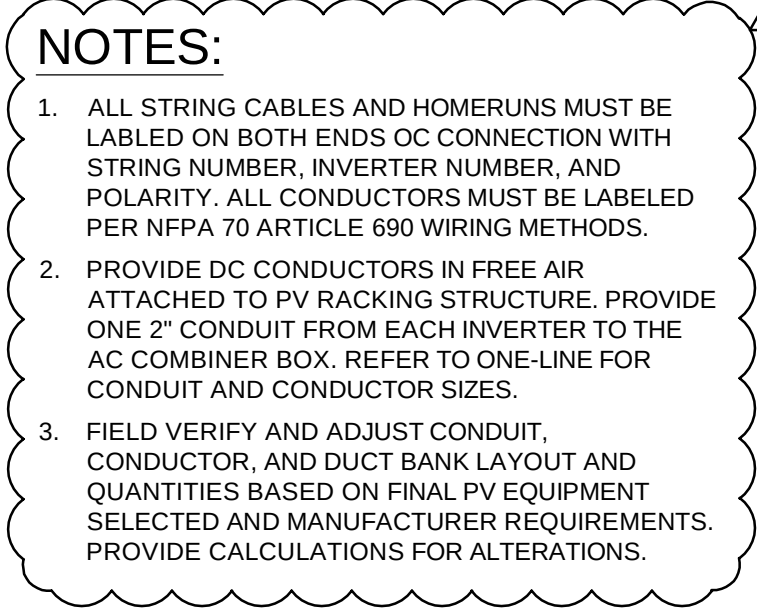
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DRAWN BY: J. ROGERS
CHECKED BY: M. SMITH
SUBMITTED BY: C. DRUMMOND
SIZE: ANSID

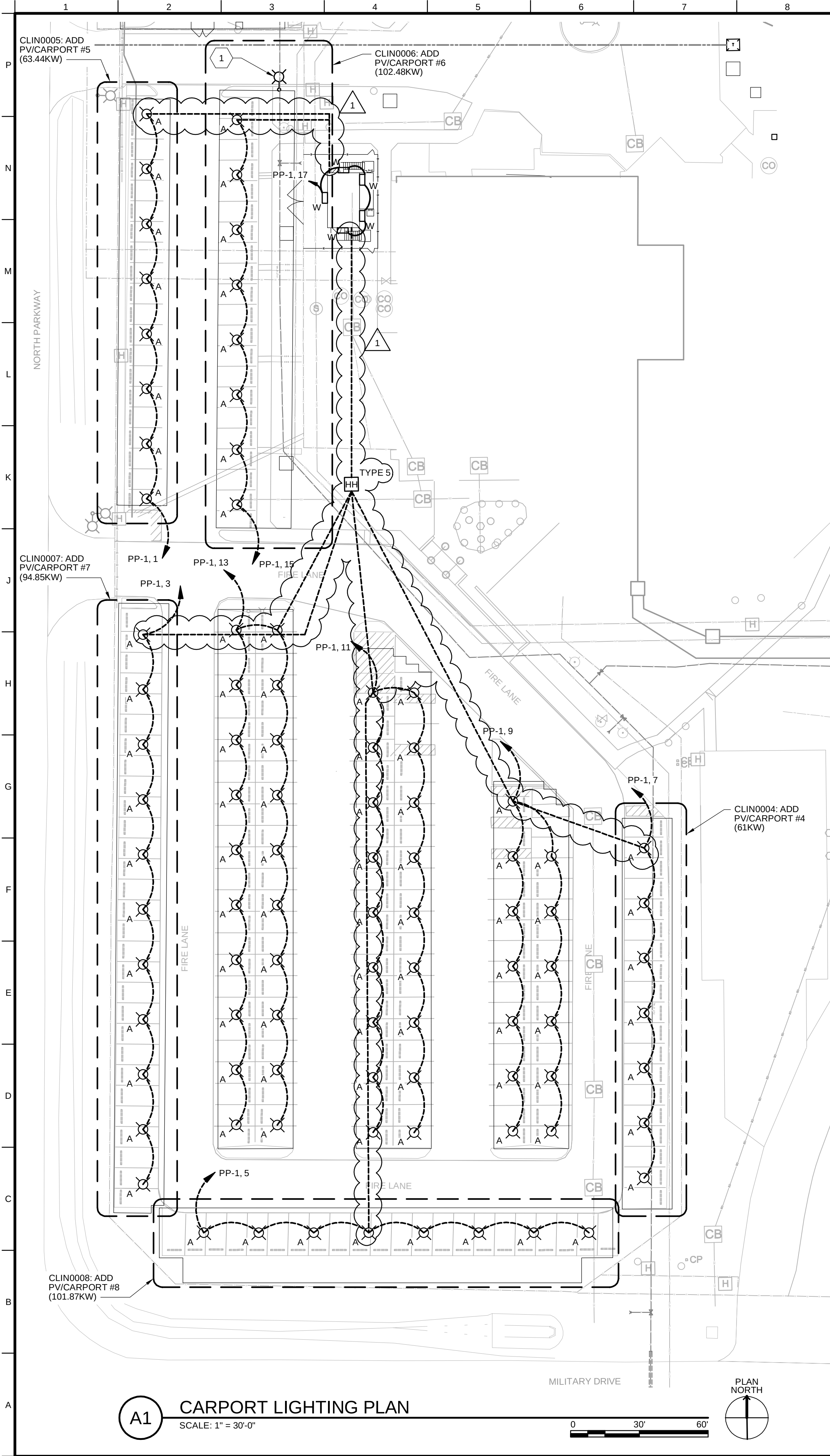
ISSUE DATE: 06 MAY 2025
SOLICITATION NO.:
CONTRACT NO.:
MICHAEL BAKER INTERNATIONAL, INC.
600 DR MARTIN LUTHER KING JR PL,
LOUISVILLE, KY 40202
MOON TOWNSHIP, PA 15108

CONROE ARMY RESERVE CENTER, CONROE, TEXAS
MICROGRID DESIGNS, 750KWDC

ELECTRICAL
ENLARGED PLANS
SHEET 2

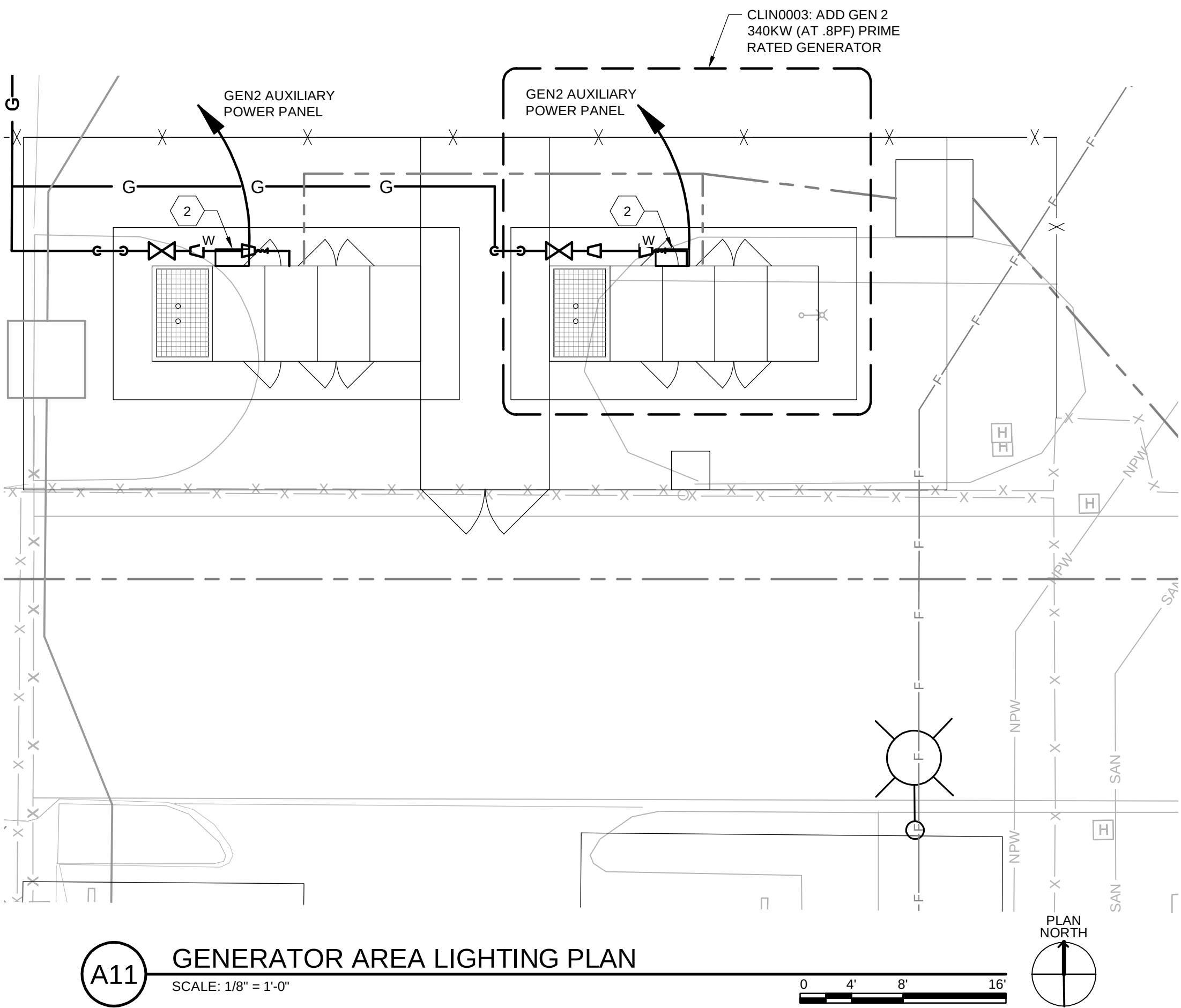
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ES401





LIGHTING FIXTURE SCHEDULE					
SYMBOL	DESCRIPTION	VOLTAGE	COLOR TEMPERATURE	LAMPS	REMARKS
A	CANOPY LIGHT, DIE CAST ALUMINUM LED,W-4 LAMPS, TYPE 5 DISTRIBUTION, IP66 RATED WITH FULL CUTOFF, MOTION DETECTOR, AND PHOTOCELL	UNIVERSAL VOLTAGE 120-277V, 60HZ	5000K	(4) - LED LAMPS	SURFACE MOUNT, TO PV ARRAY STRUCTURE. COORDINATE WITH MANUFACTURER AND PV STRUCTURE MANUFACTURER FOR MOUNTING.
W	WALL-PACK, DIE-CAST ALUMINUM HOUSING WITH HINGED, REMOVABLE DIE-CAST ALUMINUM DOOR.	UNIVERSAL VOLTAGE 120-277V, 60HZ	5000K	LED, FULL CUT OFF VISOR, IMPACT AND HEAT RESISTANT BOROSILICATE REFRACTIVE GLASS LENS	SURFACE WALL MOUNT

- KEYNOTES:**
- RELOCATE SALVAGED STREET LIGHT, POLE, AND CIRCUIT HERE. ROTATE FIXTURE 90° AND EXTEND CIRCUITS AS REQUIRED.
 - PROVIDE UNIT MOUNTED WALL PACKS ON GENERATOR ENCLOSURE. ~~POWER FROM GENERATOR AUXILIARY CIRCUIT.~~
 - REFER TO PANEL SCHEDULE ON SHEET E-700 FOR CONDUIT AND CONDUCTOR SIZES. FIELD ROUTE LIGHTING CONDUIT LAYOUT AS REQUIRED.



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AMENDMENT 002
DESCRIPTION

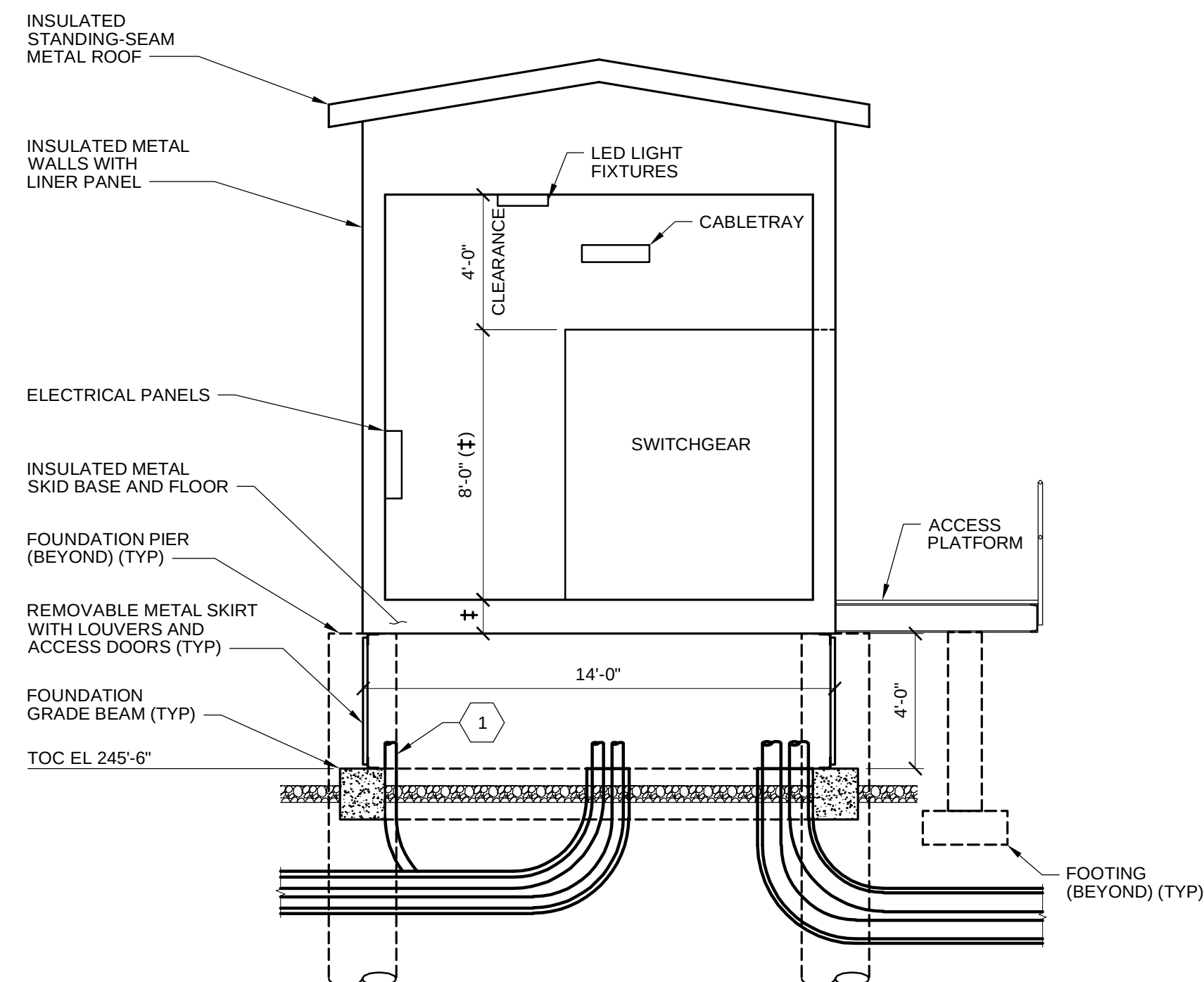
ISSUE DATE: 06 MAY 2025
SOLICITATION NO.:
CONTRACT NO.:
DESIGNED BY: M. WAGNER
DRAWN BY: J. ROGERS
CHECKED BY: M. SMITH
SUBMITTED BY: C. DRUMMOND
SIZE: ANSID

US ARMY CORPS OF ENGINEERS
LOUISVILLE DISTRICT
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MOON TOWNSHIP, PA 15108

CONROE ARMY RESERVE CENTER, CONROE, TEXAS
MICROGRID DESIGNS, 750KWDC
ELECTRICAL
SITE LIGHTING
PLANS

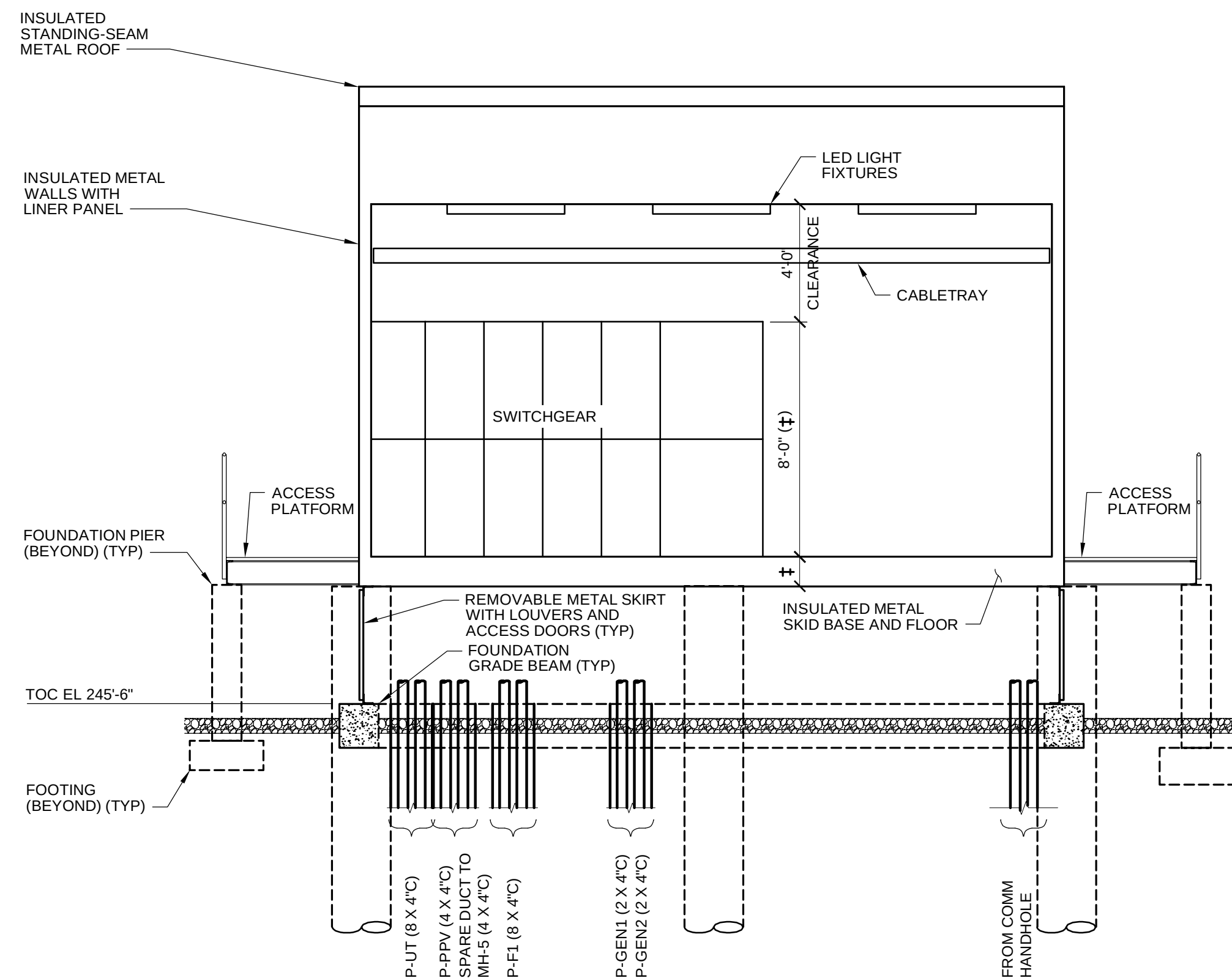
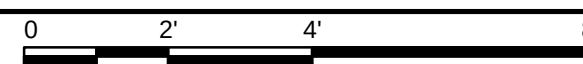
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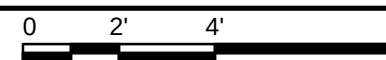
J8 SWITCHGEAR HOUSE SECTION
SCALE: 1/4" = 1'-0"

SCALE: 1/4" = 1'-0"



A8 SWITCHGEAR HOUSE SECTION

SCALE: 1/4" = 1'-0"

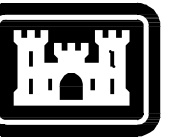


NOTES:

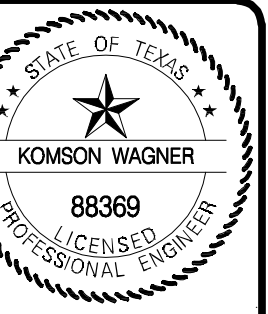
1. DESIGN, FURNISH AND INSTALL SWITCHGEAR HOUSE AND ASSOCIATED ENCLOSURE, EQUIPMENT, UTILITIES, PLATFORMS, STAIRS, LADDERS, RAILING, SKIRT AND ANCHORAGE SYSTEMS.
2. REFER TO DRAWINGS S-400 AND S-503 FOR THE SWITCHGEAR HOUSE FOUNDATION PLAN AND DETAILS.
3. REFER TO DRAWING E-106 FOR SWITCHGEAR HOUSE LAYOUT.
4. REFER TO DRAWING ES100 FOR DUCTBANK LAYOUT AND E-500 FOR DUCTBANK DETAILS.
5. CONTRACTOR SHALL ADJUST DUCTBANK/CABLE ENTRY BASED ON MANUFACTURER EQUIPMENT SELECTED.

KEYNOTES:

1. PROVIDE 2 X 4"C FROM HQ BUILDING CORNER TO CONTROL CABLE TRAY.



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600 DR MARTIN LUTHER KING JR PL, LOUISVILLE, KY 40202	DRAWN BY: J. ROGERS	CONTRACT NO.:
MICHAEL BAKER INTERNATIONAL, INC. 100 AIRSIDE DRIVE MOON TOWNSHIP, PA 15108	CHECKED BY: M. SMITH	SUBMITTED BY: S. SUMMOND
	ANSI D	

MICROGRID DESIGNS, 750KWDC

ELECTRICAL MICROGRID INTERCONNECTION SWITCHGEAR ENCLOSURE SECTIONS

HEET ID

E-107

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FEEDER CABLE AND RACEWAY SCHEDULE						
CABLE ID	FROM	TO	CABLES	RACEWAY ROUTING	RACEWAYS	CLIN
P-UT	ENTERGY SERVICE TRANSFORMER	MICROGRID INTERCONNECTION SWITCHGEAR, POI BREAKER M1	(8 SETS) 4 - 1/C, 600V, 1/C, 500 KCMIL, CU, XHHW-2	MH-7, DB-12, MH-6, DB-8	8 x 4" C	CLIN 0001
P-G1	MICROGRID INTERCONNECTION SWITCHGEAR, CIRCUIT BREAKER CB-G1	GENERATOR UNIT, G1	(2 SETS) 4 - 1/C, 600V, 500 KCMIL, CU, XHHW2	DB-11, MH-8, DB-9, MH-6, DB-8	2 x 4" C	CLIN 0001
P-G2	MICROGRID INTERCONNECTION SWITCHGEAR, CIRCUIT BREAKER CB-G2	GENERATOR UNIT, G2	(2 SETS) 4 - 1/C, 600V, 500 KCMIL, CU, XHHW2	DB-10, MH-8, DB-9, MH-6, DB-8	2 x 4" C	CLIN 0003
P-PV	MICROGRID INTERCONNECTION SWITCHGEAR, CIRCUIT BREAKER CB-PV	AC COMBINER PANEL (1200A)	(4 SETS) 4 - 1/C, 600V, 500 KCMIL, CU, XHHW2	MH-4, DB-2, MH-5, DB-1	4 x 4" C	CLIN 0001
P-F1	MICROGRID INTERCONNECTION SWITCHGEAR, CIRCUIT BREAKER CB-F1	ARC BUILDING MAIN SWITCHBOARD (MSB)	(8 SETS) 4 - 1/C, 600V, 1/C, 500 KCMIL, CU, XHHW-2	DB-13	8 x 4" C	CLIN 0001
P-F2	MICROGRID INTERCONNECTION SWITCHGEAR, CIRCUIT BREAKER CB-F2	POWER PANEL, PP-1 (600A, 480Y/277V, 3-PH, 4-W)	(2 SETS) 4 - 1/C, 600V, 500 KCMIL, CU, XHHW2	N/A	2 X 4" C	CLIN 0001
P-F3	POWER PANEL, PP-1	FIRE TRUCK STORAGE BUILDING	4 - 1/C, 600V, 500 KCMIL, CU, XHHW2	DB-14	2 x 4" C	CLIN 0009
P-001	POWER PANEL, PP-1	AUXILIARY TRANSFORMER, T-AUX (75 KVA, 480-208Y/120V, 3-PH, 4-W)	3 - 1/C, 600V, #2 AWG, CU, XHHW2, 1#6 GND	N/A	1 X 1-1/2" C	CLIN 0001
P-002	AUXILIARY TRANSFORMER, T-AUX	POWER PANEL, LP-1 (225A, 208Y/120V, 3-PH, 4-W)	4 - 1/C, 600V, #4/0AWG, CU, XHHW2, 1#4 GND	N/A	1 X 2 1/2" C	CLIN 0001
P-003	MICROGRID INTERCONNECTION SWITCHGEAR, CIRCUIT BREAKER CB-F4	MOBILE LOAD BANK CONNECTION BOX	(2 SETS) 4 - 1/C, 600V, 500 KCMIL, CU, XHHW2	N/A	2 x 4" C	CLIN 0001
P-F1	ENTERGY SERVICE TRANSFORMER (TEMPORARY FEED)	ARC BUILDING TERMINATION BOX (TEMPORARY FEED)	(8 SETS) 4 - 1/C, 600V, 1/C, 500 KCMIL, CU, XHHW-2	DB-7	8 x 4" C	CLIN 0001

SEE POWER PANEL SCHEDULE FOR BRANCH CIRCUITS WIRING REQUIREMENTS

SWITCHGEAR'S MULTIFUNCTION RELAY PROTECTION FUNCTION REQUIREMENTS & GENERATOR PROTECTION RELAY'S AND INVERTER'S PROTECTION FUNCTION REQUIREMENTS

SWITCHGEAR MULTIFUNCTION RELAY	MINIMUM PROTECTIVE RELAY DEVICE FUNCTIONS PER ANSI STANDARD
MFR-M1 (MAIN BRK.)	25, 27, 32, 51, 51N, 59, 81
MFR-TIE (TIE BREAKER)	50, 51
MFR-PV (PV IN. BRK.)	25, 27, 32, 51, 51N, 59, 81
MFR-G1 (GENERATOR BRK.)	25, 27, 32, 40, 46, 50, 51V, 51G, 59, 81
MFR-G2 (GENERATOR BRK.)	25, 27, 32, 40, 46, 50, 51V, 51G, 59, 81
MFR-F1 (FEEDER BREAKER)	27, 50, 51
MFR-F1A (FEEDER BREAKER)	27, 50, 51
MFR-F2 (FEEDER BREAKER)	27, 50, 51
MFR-F2A (FEEDER BREAKER)	27, 50, 51
MFR-F4 (FEEDER BREAKER)	27, 50, 51
MFR-SP1 (SPARE BREAKER)	25, 27, 47, 50, 51, 59, 81
MFR-SP2 (SPARE BREAKER)	25, 27, 47, 50, 51, 59, 81
MFR-SP3 (SPARE BREAKER)	25, 27, 47, 50, 51, 59, 81

GENERATOR RELAY FUNCTIONS	25, 27, 32, 46, 47, 50, 51V, 51G, 59, 81
PV INVERTER (BUILT-IN PROTECTIVE FUNCTIONS)	25, 27, 47, 50, 51, 59, 81

NOTES:

VTs AND CTs INSTRUMENTS BY EQUIPMENT MANUFACTURER.

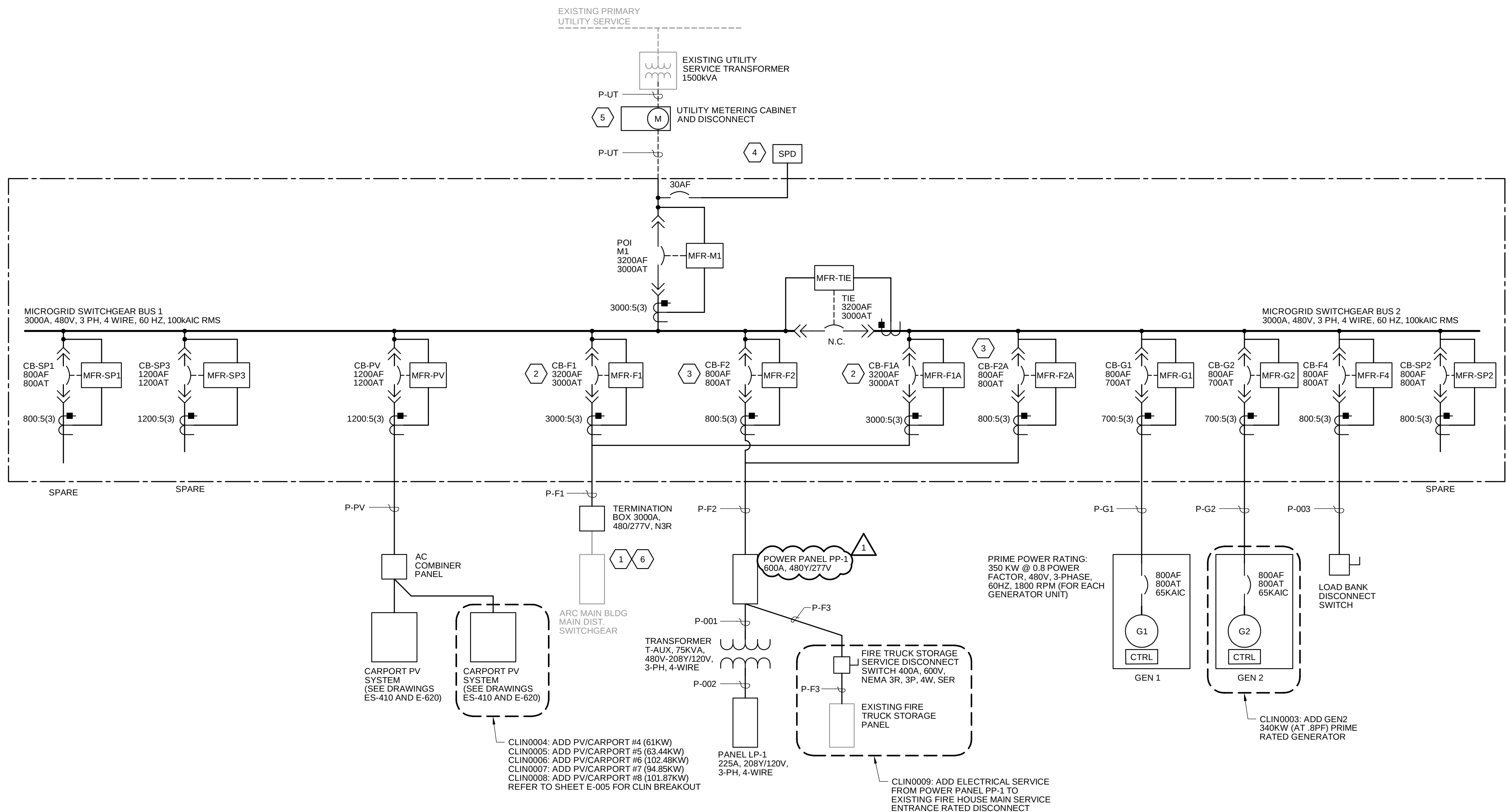
BREAKER STATUS, TRIP, CLOSE, BLOCK CLOSE BY MANUFACTURER AS REQUIRED TO PERFORM MICROGRID MONITORING AND CONTROL FUNCTIONS.

NOTES:

1. SEE SHEET E-005 FOR CLIN DESCRIPTION

KEYNOTES:

1. CONTRACTOR IS RESPONSIBLE FOR COORDINATING OUTAGES WITH UTILITY COMPANY. PROVIDE TEMPORARY POWER FOR ALL ASSOCIATED LOADS AS REQUIRED TO MAINTAIN ARC BUILDING OPERATIONS DURING PROJECT CONSTRUCTION. COORDINATE ALL DISRUPTIONS WITH OWNER PRIOR TO ANY OUTAGES.
2. CONNECTION AND TERMINATION BETWEEN BREAKERS CB-F1 AND CB-F1A BY SWITCHGEAR MANUFACTURER.
3. CONNECTION AND TERMINATION BETWEEN BREAKERS CB-F2 AND CB-F2A BY SWITCHGEAR MANUFACTURER.
4. PROVIDE SPD AHEAD OF POI BREAKER. PROVIDE BREAKER AND CONDUCTOR SIZE PER MANUFACTURER REQUIREMENTS.
5. PROVIDE 3000A, 480Y/277V, 65 KAIC, NEMA 3R LOAD BREAK DISCONNECT, METER, AND ASSOCIATED INSTRUMENTATION (VT,CT). ELECTRICAL EQUIPMENT AND INSTALLATION MUST COMPLY WITH ENERGY DISTRIBUTED ENERGY RESOURCE STANDARDS FOR DISTRIBUTION INTERCONNECTION AND CUSTOMER INSTALLATION STANDARDS FOR ELECTRIC SERVICE MANUAL.
6. PROVIDE 3000A, 480Y/277V, 65KAIC, NEMA 3R TERMINATION BOX WITH CT'S, DOOR BARRIERS, AND GROUND BUS.



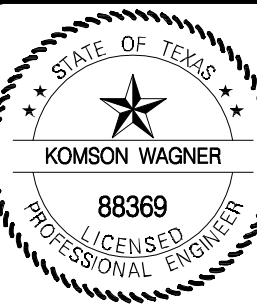
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MICROGRID INTERCONNECTION SWITCHGEAR ONE-LINE DIAGRAM

SCALE: NTS



**US Army Corps
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US ARMY CORPS OF ENGINEERS 600 DR MARTIN LUTHER KING JR PL LOUISVILLE, KY 40202	DESIGNED BY:	M. WAGNER	ISSUE DATE:	06 MAY 2025
	DRAWN BY:	J. ROGERS	SOLICITATION NO.:	
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	SUBMITTED BY:	C. DRUMMOND		
MICHAEL BAKER INTERNATIONAL, INC. 100 AIRSIDE DRIVE MOONT TOWNSHIP, PA 15108				

CONROE ARMY RESERVE CENTER, CONROE, TEXAS
MICROGRID DESIGNS. 750KWDC

ELECTRICAL MICROGRID INTERCONNECTION SWITCHGEAR ONE-LINE DIAGRAM

SHEET ID

E-600

